Joel Mackenzie

Brisbane, Queensland, Australia
https://jmmackenzie.io
joel.mackenzie@uq.edu.au

Education

Doctorate of Philosophy (Computer Science). February 2016 – October 2019
PhD Thesis: *Managing Tail Latency in Large Scale Information Retrieval Systems.*
RMIT University, Melbourne, Australia.

Bachelor of Computer Science, Hons (First Class). February 2012 – November 2015
Honours Thesis: *Efficiency-Effectiveness Trade-offs for Location Aware Search.*
RMIT University, Melbourne, Australia.

Academic Positions

The University of Queensland – Senior Lecturer January 2026 – Ongoing

The University of Queensland – Lecturer January 2022 – December 2025
Teaching and Research in Computer Science, focusing on Information Retrieval, Data Structures and Algorithms.

The University of Melbourne – Postdoctoral Research Fellow February 2020 – January 2022
Worked with Prof. Alistair Moffat on the ARC Discovery Project *Efficient Compression and Querying Techniques for Massive Text Collections.*

Teaching Experience

The University of Queensland

- Algorithms & Data Structures – Course coordinator and Lecturer. 2023–2025
- Concurrency: Theory & Practice – Course coordinator and Co-Lecturer. 2025
- Advanced Database Systems – Course coordinator and Lecturer. 2022
- Introduction to Data Science – Course coordinator and Co-Lecturer. 2023

Research Interests

I am primarily interested in the efficient data structures and algorithms, and how they apply to practical problems. Most of my work has focused on the discipline of Information Retrieval, including efficient indexing, retrieval, and ranking. I am also interested in the evaluation of IR systems, and understanding user behavior.

Publications

Z. Li and J. Mackenzie: **Simple Techniques for Efficient Top-k Batch Query Processing.** In *Information Retrieval Research Journal*, April, 2026.

B. Rule and J. Mackenzie: **Fast, Compact, Immediate-Access Indexing for Learned Sparse Retrieval Systems.** In *Proceedings of the 48th European Conference on Information Retrieval (ECIR 2026)*, March, 2026.

W. Mansour, J. Shane Culpepper, A. Yates, and J. Mackenzie: **Revisiting Human-vs-LLM judgments**

using the TREC Podcast Track. In *Proceedings of the 48th European Conference on Information Retrieval (ECIR 2026)*, March, 2026.

J. Mackenzie, S. MacAvaney, A. Mallia, and M. Siedlaczek: **Practical, Efficient, In-Memory Inverted Indexes.** In *Proceedings of the 48th European Conference on Information Retrieval (ECIR 2026)*, March, 2026.

T. Chen, H. Chen, J. Mackenzie, F. Zhu, L. Pizzato, R. Ng, and A. Leontjeva: **Workshop on Agentic AI in Financial Services.** In *Proceedings of the 40th Annual AAAI Conference on Artificial Intelligence (AAAI 2026)*, January, 2026.

F. Ariai, J. Mackenzie, and G. Demartini: **Natural Language Processing for the Legal Domain: A Survey of Tasks, Datasets, Models, and Challenges.** In *ACM Computing Surveys*, December, 2025.

L. Gallagher, J. Mackenzie, and A. Moffat: **Empirical Asymptotic Growth of Dynamic Pruning Mechanisms.** In *Proceedings of the 3rd Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region (SIGIR-AP 2025)*, December, 2025.

A. Moffat, J. Mackenzie, A. Mallia, and M. Petri: **A Flexible Resource for Top-Weighted Comparisons Between Sets and Rankings.** In *Proceedings of the 48th International ACM Conference on Research and Development in Information Retrieval (SIGIR 2025)*, July, 2025.

J. Mackenzie, S. MacAvaney, A. Mallia, and M. Siedlaczek: **Efficient In-Memory Inverted Indexes: Theory and Practice.** In *Proceedings of the 48th International ACM Conference on Research and Development in Information Retrieval (SIGIR 2025)*, July, 2025.

L. Dunn, L. Gallagher, and J. Mackenzie: **Approximate Bag-of-Words Top-k Corpus Graphs.** In *Proceedings of the 47th European Conference on Information Retrieval (ECIR 2025)*, April, 2025.

W. Mansour, J. Shane Culpepper, and J. Mackenzie: **Examining the Impact of Transcript Accuracy on Podcast Search and Re-Ranking.** In *Proceedings of the 47th European Conference on Information Retrieval (ECIR 2025)*, April, 2025.

J. Alanko, E. Biagi, J. Mackenzie, and S. Puglisi: **Batched k-mer lookup on the Spectral Burrows-Wheeler Transform.** In the *2025 Proceedings of the Symposium on Algorithm Engineering and Experiments (ALENEX 2025)*, January, 2025.

A. Moffat, J. Mackenzie, A. Mallia, and M. Petri: **Rank-Biased Quality Measurement for Sets and Rankings.** In *Proceedings of the 2nd Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region (SIGIR-AP 2024)*, December, 2024.

J. Trippas, L. Gallagher, and J. Mackenzie: **Re-evaluating the Command-and-Control Paradigm in Conversational Search Interactions.** In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM 2024)*, October, 2024.

W. Mansour, S. Zhuang, G. Zuccon, and J. Mackenzie: **Revisiting Document Expansion and Filtering for Effective First-Stage Retrieval.** In *Proceedings of the 47th International ACM Conference on Research and Development in Information Retrieval (SIGIR 2024)*, July, 2024.

J. Trippas, S. F. D. Al Lawati, J. Mackenzie, and L. Gallagher: **What do Users Really Ask Large Language Models? An Initial Log Analysis of Google Bard Interactions in the Wild.** In *Proceedings of the 47th International ACM Conference on Research and Development in Information Retrieval (SIGIR 2024)*, July, 2024.

M. Fröbe, J. Mackenzie, B. Mitra, F. M. Nardini, and M. Potthast: **ReNeuIR at SIGIR 2024: The Third**

Workshop on Reaching Efficiency in Neural Information Retrieval. In *Proceedings of the 47th International ACM Conference on Research and Development in Information Retrieval (SIGIR 2024)*, July, 2024.

A. Moffat and J. Mackenzie: **How Much Freedom Does An Effectiveness Metric Really Have?** In *Journal of the Association for Information Science and Technology (JASIST)*, January, 2024.

J. Mackenzie and A. Moffat: **Lossy Compression Options for Dense Index Retention.** In *Proceedings of the 1st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region (SIGIR-AP 2023)*, November, 2023.

J. Mackenzie, S. Zhuang, and G. Zuccon: **Exploring the Representation Power of SPLADE Models.** In *Proceedings of the 2023 ACM SIGIR International Conference on the Theory of Information Retrieval (ICTIR 2023)*, July, 2023.

Z. Li and J. Mackenzie: **Profiling and Visualizing Dynamic Pruning Algorithms.** In *Proceedings of the 46th International ACM Conference on Research and Development in Information Retrieval (SIGIR 2023)*, July, 2023.

S. Bruch, J. Mackenzie, M. Maistro, and F. M. Nardini: **ReNeuIR at SIGIR 2023: The Second Workshop on Reaching Efficiency in Neural Information Retrieval.** In *Proceedings of the 46th International ACM Conference on Research and Development in Information Retrieval (SIGIR 2023)*, July, 2023.

J. Mackenzie and A. Moffat: **Index-Based Batch Query Processing Revisited.** In *Proceedings of the 45th European Conference on Information Retrieval (ECIR 2023)*, April, 2023.

A. Moffat and J. Mackenzie: **Efficient Immediate-Access Dynamic Indexing.** In *Information Processing and Management (IPM)*, 2023.

J. Mackenzie and A. Trotman and J. Lin: **Efficient Document-at-a-Time and Score-at-a-Time Query Evaluation for Learned Sparse Representations.** In *ACM Transactions on Information Systems (TOIS)*, 2022.

A. Moffat and J. Mackenzie: **Immediate-Access Indexing Using Space-Efficient Extensible Arrays.** In *Proceedings of the 26th Australasian Document Computing Symposium (ADCS 2022)*, December, 2022.

J. Mackenzie, A. Mallia, A. Moffat, and M. Petri: **Accelerating Learned Sparse Indexes Via Term Impact Decomposition.** In *Findings of the Association for Computational Linguistics: EMNLP 2022 (EMNLP Findings 2022)*, December, 2022.

J. Mackenzie, M. Petri, and A. Moffat: **Tradeoff Options for Bipartite Graph Partitioning.** In *IEEE Transactions on Data Engineering (TKDE)*, 2022.

J. Mackenzie, M. Petri, and L. Gallagher: **IOQP: A simple Impact-Ordered Query Processor written in Rust.** In *Proceedings of the 3rd International Conference on Design of Experimental Search and Information Retrieval Systems (DESIRES 2022)*, August, 2022.

D. La Barbera, K. Roitero, J. Mackenzie, D. Spina, G. Demartini, and S. Mizzaro: **Brisbane–Udine–Melbourne at CLEF2022 CheckThat! Lab: A Composite Deep Learning Approach to Fake News Detection using Evidence Retrieval.** In *Working Notes of CLEF 2022 - Conference and Labs of the Evaluation Forum (CLEF 2022)*, September, 2022.

A. Moffat, J. Mackenzie, P. Thomas, and L. Azzopardi: **A Flexible Framework for Offline Effectiveness Metrics.** In *Proceedings of the 45th International ACM Conference on Research and Development in Information Retrieval (SIGIR 2022)*, July, 2022.

A. Mallia, J. Mackenzie, T. Suel, and N. Tonellotto: **Faster Learned Sparse Retrieval with Guided Traversal.** In *Proceedings of the 45th International ACM Conference on Research and Development in Information Retrieval (SIGIR 2022)*, July, 2022.

A. Trotman, J. Mackenzie, P. Parameswaran, and J. Lin: **A Common Framework for Exploring Document-at-a-Time and Score-at-a-Time Retrieval Methods.** In *Proceedings of the 45th International ACM Conference on Research and Development in Information Retrieval (SIGIR 2022)*, July, 2022.

J. Mackenzie and D. Spina: **Report on the 25th Australasian Document Computing Symposium (ADCS 2021).** In *SIGIR Forum*, 2022. J. Mackenzie, M. Petri, and A. Moffat: **Efficient Query Processing Techniques for Next-Page Retrieval.** In *Information Retrieval Journal (IRJ)*, 2022.

J. Mackenzie, M. Petri, and A. Moffat: **Anytime Ranking on Document-Ordered Indexes.** In *ACM Transactions on Information Systems (TOIS)*, 2022.

A. Moffat and J. Mackenzie: **Conferences, Journals, Preprints, and Reviewer Expectations.** In *SIGIR Forum*, 2021.

J. Mackenzie and A. Moffat: **Cost-Effective Updating of Distributed Reordered Indexes.** In *Proceedings of the 25th Australasian Document Computing Symposium (ADCS 2021)*, December, 2021. Best paper award.

J. Mackenzie, M. Petri, and A. Moffat: **A Sensitivity Analysis of the MSMARCO Collection.** Unpublished technical report, 2021. Available: `https://arxiv.org/abs/2112.03396`.

J. Lin, X. Ma, J. Mackenzie, and A. Mallia: **On the Separation of Logical and Physical Ranking Models for Text Retrieval Applications.** In *Proceedings of the Second Biennial Conference on Design of Experimental Search and Information Retrieval Systems (DESIRES 2021)*, September, 2021.

J. Mackenzie, M. Petri, and A. Moffat: **Faster Index Reordering with Bipartite Graph Partitioning.** In *Proceedings of the 44th International ACM Conference on Research and Development in Information Retrieval (SIGIR 2021)*, July, 2021.

L. Azzopardi, J. Mackenzie, and A. Moffat: **ERR is not C/W/L: Exploring the Relationship Between Expected Reciprocal Rank and Other Metrics.** In *Proceedings of the 2021 ACM SIGIR International Conference on the Theory of Information Retrieval (ICTIR 2021)*, July, 2021. Best paper award.

J. Mackenzie and A. Moffat: **Modality Effects When Simulating User Querying Tasks.** In *Proceedings of the 2021 ACM SIGIR International Conference on the Theory of Information Retrieval (ICTIR 2021)*, July, 2021.

R. Benham, J. Mackenzie, J. S. Culpepper, and A. Moffat: **Different Keystrokes for Different Folks: Visualizing Crowdworker Querying Behavior.** In *Proceedings of the 6th ACM SIGIR Conference on Human Information Interaction and Retrieval (CHIIR 2021)*, March, 2021.

J. Mackenzie and A. Moffat: **Examining the Additivity of Top-k Query Processing Innovations.** In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM 2020)*, October, 2020.

J. Mackenzie, R. Benham, M. Petri, J. R. Trippas, J. S. Culpepper, and A. Moffat: **CC-News-En: A Large English News Corpus** In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM 2020)*, October, 2020.

J. Mackenzie, Z. Dai, L. Gallagher, and J. Callan: **Efficiency Implications of Term Weighting for Pas-**

sage Retrieval. In *Proceedings of the 43rd International ACM Conference on Research and Development in Information Retrieval (SIGIR 2020)*, July, 2020.

J. Lin, J. Mackenzie, C. Kamphuis, C. Macdonald, A. Mallia, M. Siedlaczek, A. Trotman, and A. de Vries: **Supporting Interoperability Between Open-Source Search Engines with the Common Index File Format.** In *Proceedings of the 43rd International ACM Conference on Research and Development in Information Retrieval (SIGIR 2020)*, July, 2020.

R. Benham, J. Mackenzie, A. Moffat, and J. S. Culpepper: **Boosting Search Performance Using Query Variations.** In *ACM Transactions on Information Systems (TOIS)*, October, 2019.

A. Mallia, M. Siedlaczek, J. Mackenzie, and T. Suel: **PISA: Performant Indexes and Search for Academia.** In *Proceedings of the Open-Source IR Replicability Challenge (OSIRRC 2019)*, July, 2019.

M. Petri, A. Moffat, J. Mackenzie, J. S. Culpepper, and D. Beck: **Accelerated Query Processing Via Similarity Score Prediction.** In *Proceedings of the 42nd International ACM Conference on Research and Development in Information Retrieval (SIGIR 2019)*, July, 2019.

J. Mackenzie, K. Gupta, F. Qiao, A. H. Awadallah, and M. Shokouhi: **Exploring User Behavior in Email Re-Finding Tasks.** In *Proceedings of the 2019 World Wide Web Conference (WWW 2019)*, May, 2019.

J. Mackenzie, A. Mallia, M. Petri, J. S. Culpepper, and T. Suel: **Compressing Inverted Indexes with Recursive Graph Bisection: A Reproducibility Study.** In *Proceedings of the 41st European Conference on Information Retrieval (ECIR 2019)*, April, 2019.

L. Gallagher, J. Mackenzie, and J. S. Culpepper: **Revisiting Spam Filtering in Web Search.** In *Proceedings of the 23rd Annual Australasian Document Computing Symposium (ADCS 2018)*, December, 2018.

R. Benham, L. Gallagher, J. Mackenzie, B. Liu, X. Lu, F. Scholer, A. Moffat, and J. S. Culpepper: **RMIT at the 2018 TREC CORE Track.** In *Proceedings of the 27th Text REtrieval Conference (TREC 2018)*, November, 2018.

R. Benham, J. S. Culpepper, L. Gallagher, X. Lu, and J. Mackenzie : **Towards Efficient and Effective Query Variant Generation.** In *Proceedings of the First Biennial Conference on Design of Experimental Search and Information Retrieval Systems (DESIRES 2018)*, August, 2018.

J. Mackenzie, C. Macdonald, F. Scholer, and J. S. Culpepper: **On the Cost of Negation for Dynamic Pruning.** In *Proceedings of the 40th European Conference on Information Retrieval (ECIR 2018)*, March, 2018.

J. Mackenzie, J. S. Culpepper, R. Blanco, M. Crane, C. L. A. Clarke, and J. Lin: **Query Driven Algorithm Selection in Early Stage Retrieval.** In *Proceedings of the 11th International ACM Conference on Web Search and Data Mining (WSDM 2018)*, February, 2018.

J. Mackenzie, F. Scholer, and J. S. Culpepper: **Early Termination Heuristics for Score-at-a-Time Index Traversal.** In *Proceedings of the 22nd Annual Australasian Document Computing Symposium (ADCS 2017)*, December, 2017.

L. Gallagher, J. Mackenzie, R. Benham, R-C. Chen, F. Scholer, and J. S. Culpepper: **RMIT at the NTCIR-13 We Want Web Task.** In *Proceedings of the 13th NTCIR Conference on Evaluation of Information Access Technologies (NTCIR 2017)*, December, 2017.

R. Benham, L. Gallagher, J. Mackenzie, T. T. Damessie, R-C. Chen, F. Scholer, A. Moffat, and J. S. Culpepper: **RMIT at the 2017 TREC CORE Track.** In *Proceedings of the 26th Text REtrieval Conference (TREC 2017)*, November, 2017.

J. Mackenzie: **Managing Tail Latencies in Large Scale IR Systems.** In *Proceedings of the 40th International ACM Conference on Research and Development in Information Retrieval (SIGIR 2017)*, August, 2017.

M. Crane, J. S. Culpepper, J. Lin, J. Mackenzie, and A. Trotman: **A Comparison of Document-at-a-Time and Score-at-a-Time Query Evaluation.** In *Proceedings of the 10th International ACM Conference on Web Search and Data Mining (WSDM 2017)*, February, 2017.

J. Mackenzie, R-C. Chen, and J. S. Culpepper: **RMIT at TREC 2016 LiveQA Track.** In *Proceedings of the 25th Text REtrieval Conference (TREC 2016)*, November, 2016.

J. Mackenzie, F. M. Choudhury, and J. S. Culpepper: **Efficient Location-Aware Web Search.** In *Proceedings of the 20th Annual Australasian Document Computing Symposium (ADCS 2015)*, December, 2015. Best paper award.

Publicity

Amazon Science Blog: Article titled *"More efficient and reliable retrieval of distributed data"* features our 2021 ACM TOIS publication *"Anytime Ranking on Document-Ordered Indexes."*

Awards and Funding

Australian Research Council Discovery Early Career Researcher Award (2026–2029): $500,000.
Sustainable Mineral Trailblazer Seed Funding (2025): $20,000.
Google Research Scholar Award (2024): $60,000 (USD).
The University of Queensland New Staff Research Grant (2022): $50,000.
The University of Melbourne Early Career Seed Funding (2021): $20,000.
Best Paper Award: ADCS 2021, ICTIR 2021, ADCS 2017 (Runner up), ADCS 2015.
RMIT CIDDA CReD Research Excellence Award: For *"Exploring User Behavior in Email Re-Finding Tasks"* (WWW 2019).
RMIT TIGER/BDDA Group Student Paper Award: For *"Query Driven Algorithm Selection in Early Stage Retrieval"* (WSDM 2018).
SIGIR Travel Award: SIGIR 2017, WSDM 2017.
Australian Postgraduate Award: PhD scholarship from 2016 to 2019.
RMIT Vice Chancellors List Award (Top 2% of higher education university-wide): 2014, 2015.
Outstanding Reviewer/Program Chair Award: SIGIR 2024, CIKM 2024, SIGIR-AP 2024.
Commendation for Excellence in Student Learning: EAIT Faculty, University of Queensland, 2024.
Research Excellence Award: Open Science Practices: EAIT Faculty, University of Queensland, 2024. Awarded to the entire IELab group.

Service

General Chair
Australasian Document Computing Symposium (ADCS) 2021.

Program Chair
Australasian Document Computing Symposium (ADCS) 2022.

Short Papers Chair
International ACM Conference on Research and Development in Information Retrieval (SIGIR) 2025.

Resource Papers Chair
International ACM Conference on Research and Development in Information Retrieval (SIGIR) 2026.

Demo Chair
International ACM Conference on Research and Development in Information Retrieval (SIGIR) 2024.

Proceedings Chair

International ACM SIGIR Conference on Information Retrieval in the Asia Pacific (SIGIR-AP) 2023.
ACM SIGIR Conference on Human Information Interaction and Retrieval (CHIIR) 2021 and 2025.
International ACM Conference on Web Search and Data Mining (WSDM) 2019.

SIGIR Artifact Evaluation Committee (AEC)
Member: 2021–Present.

Area Chair
International Conference on Computational Linguistics (COLING) 2022.

Senior Program Committee Member
International ACM Conference on Information and Knowledge Management (CIKM): 2024 and 2025.

International ACM Conference on Research and Development in Information Retrieval (SIGIR): 2026.

European Conference on Information Retrieval (ECIR): 2026.

Program Committee Member or Reviewer
Australasian Document Computing Symposium (ADCS): 2016–2019.
SIAM Symposium on Algorithm Engineering and Experiments (ALENEX): 2026.
International ACM Conference on Information and Knowledge Management (CIKM): 2016–2023.
European Conference on Information Retrieval (ECIR): 2021–2026.
Empirical Methods in Natural Language Processing (EMNLP): 2020–2022.
The ACM SIGIR International Conference on Innovative Concepts and Theories in Information Retrieval (ICTIR): 2025.
The Journal of Information Science: 2019.
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD): 2021.
International ACM Conference on Research and Development in Information Retrieval (SIGIR): 2019–2024.
International ACM SIGIR Conference on Information Retrieval in the Asia Pacific (SIGIR-AP) 2024–2025.
International Symposium on String Processing and Information Retrieval (SPIRE): 2019, 2022.
IEEE Transactions on Knowledge and Data Engineering (TKDE): 2022.
ACM Transaction on Information Systems (TOIS): 2020.
The International Journal on Very Large Databases (VLDB): 2017.
Workshop on Insights from Negative Results in NLP: 2022 (at ACL).
International ACM Conference on Web Search and Data Mining (WSDM): 2021–2023.
The World Wide Web Conference (TheWebConf, *aka* WWW): 2022–2025.

I also review grants for the Australian Research Council.